

Python OCR & Document Automation

Custom Software Development & Intelligent Data Extraction Solutions

CORE TECHNOLOGY STACK

Python

Tesseract OCR

OpenCV

pdf2image

OpenPyXL

Tkinter GUI

Regular Expressions (Regex)

MODULAR PIPELINE ARCHITECTURE

main.py — List Orchestrator



ocr_utils.py — Converts PDF pages & Scans to text via OpenCV + Tesseract



extrator_empenho.py

Commitment Data Extraction

extrator_fiscal.py

Fiscal Data & Invoices

extrator_bancario.py

Bank Identification & Slips



planilha_utils.py — Structures & Writes Clean Data Directly to Excel



renomear_utils.py — Auto-renames PDFs & manages batch queue

⌚ When finished, lists are cleared and the next PDF batch begins automatically.

VERSATILE AUTOMATION USE CASES

While originally designed for high-volume banking and public sector vouchers, this modular Python architecture can be tailored to solve diverse document ingestion challenges across multiple industries:

Industry / Domain	Target Documents	Extracted Fields
Accounting & Finance	Bank slips, payment vouchers, corporate expense receipts, invoices.	Dates, amounts, bank codes, beneficiary name, transaction IDs.
Public Sector & Legal	Administrative commitment orders (*empenhos*), official notices, certificates.	Document numbers, government agency, fiscal values, timestamps.
Supply Chain & Logistics	Bills of lading, delivery receipts, warehouse inventory sheets, packing lists.	Tracking codes, weights, item descriptions, recipient names.
Scanned & Legacy Archives	Non-searchable PDF scans, low-quality photos, historical paper records.	Custom regex patterns mapped to structured Excel or database tables.

KEY CLIENT BENEFITS

**100% Elimination of Manual Entry**

Stop wasting hundreds of hours per month typing data line-by-line. Automate batch processing with a single click.

**Advanced OCR & Preprocessing**

OpenCV image correction (perspective, thresholding, noise removal) guarantees high accuracy even on poor-quality scans.

**Custom Desktop App Interface**

Built with Tkinter for a clean, user-friendly GUI. No programming skills required for end-users to operate the tool locally.

**Ready-to-Use Excel Output**

Data is instantly structured, validated, and exported via OpenPyXL, completely organized for auditing and archiving.